AI33-MPOPT: Garkness Geometry and Information-Modulated Cosmology

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Abstract

This work presents AI33-MPOPT, a unified cosmological framework integrating six interacting mathematical layers: (1) local FLRW dynamics, (2) vacuum-tension geometry via the Garkness field, (3) nonlocal correlation structure encoded by the Ξ-kernel, (4) a geometric information functional imposing global consistency conditions, (5) a spectral layer defined by the Rivero–Zeta function, and (6) a 33-sector geometric decomposition. These components combine into a single governing dynamical law—the AI33-MPOPT Master Equation—derived from an information-augmented action principle. The framework predicts a stable, information-selected attractor characterized by constant vacuum-tension, frozen spectral invariants, and a sector-resolved nonlocal correlation structure. The resulting cosmology exhibits a non-monotonic effective equation of state, a smoother transition to late-time acceleration, scale-dependent growth suppression, and sector-dependent correlation signatures. Mathematical consistency is established through proofs of well-posedness, boundedness, spectral continuity, and preservation of sector structure. The AI33-MPOPT architecture provides a coherent, extensible, and testable foundation for information-modulated cosmology, unifying local, nonlocal, spectral, and informational physics within a single operational framework. This work is intended as a formal theoretical framework and does not assert direct empirical validation, but rather establishes a mathematically consistent structure within which cosmological behaviors may be analyzed.

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1. Information Geometry of Garkness

This section establishes the geometric, informational, and structural layer between the global action (Super-Duper Formula), the spectral invariant layer (Rivero–Zeta), and the reduced cosmological dynamics (FLRW + scalar + pause-factor).

1.1 Mathematical Classification

The AI33-MPOPT system is a constrained, semilinear, nonlocal integro-differential cosmological model. It is not modified gravity, not scalar–tensor gravity, and not a phenomenological ΛCDM extension. The geometric information functional I_{geom} acts as a constraint selector on admissible solutions, not as a dynamical action.

1.2 The Garkness Geometric State

Definition 1.1 (Geometric State). *The geometric state of the vacuum is defined as:*

$$\mathcal{S}_{\text{geom}} = \big(G(x),\, \Xi(x,y),\, p(x);\, \sigma\big) \tag{1}$$

with components:

- $G(x)$*: vacuum-tension geometric scalar field (not a particle field);*
- $\Xi(x,y)$*: nonlocal geometric correlation kernel;*
- $p(x)$*: pause-factor modulation field;*
- $\sigma \in \{1, 2, \ldots, 33\}$*: sector label (solution-space decomposition index).*

1.3 The Geometric Information Functional

Definition 1.2 (Information Functional). *The geometric information functional is defined as:*

$$I_{\text{geom}} = I\big[G(x),\, \Xi(x,y),\, p(x);\, \sigma\big] \tag{2}$$

satisfying:

(i) Basis independence under matter/light basis changes;

(ii) Nonlocality via global dependence on $\Xi(x,y)$;

(iii) Causal safety (non-signaling);

(iv) Physical constraint on dynamical solutions.

The functional I_{geom} is a constraint selector, not a dynamical action. It restricts the space of admissible cosmological trajectories.

1.4 The Ξ-Kernel: Nonlocal Correlation Structure

Definition 1.3 (Ξ-Kernel Properties). *The kernel $\Xi : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ is a symmetric, positive, integrable nonlocal operator satisfying:*

$$\Xi(x,y) = \Xi(y,x) \quad \textit{(symmetry)} \tag{3}$$

$$\Xi(x,y) \geq 0 \quad \textit{(positivity)} \tag{4}$$

$$\int_{\mathcal{M}} |\Xi(x,y)|\, d^4y < \infty \quad \textit{(integrability)} \tag{5}$$

The Ξ-kernel is non-signaling and causally safe. It acts as a geometric correlation constraint, not a propagating field, and does not introduce superluminal effects.

1.5 Sector Label σ and the 33-Web

The sector label σ indexes the geometric sectors: $\sigma \in \{1, 2, \ldots, 33\}$. Each σ corresponds to a distinct geometric sector, correlation structure, and asymptotic behavior.

The sector labels are solution-space decomposition indices, not physical universes. The 33-web is a block decomposition of the Ξ-kernel and a partition of correlation structure inside a single universe.

1.6 Relation to the Rivero–Zeta

The Rivero–Zeta spectral invariant is defined on the spectral operator:

$$\mathcal{D}(t) = -\Delta_{g(t)} + M^2 \tag{6}$$

The spectral operator depends smoothly on the background geometry. The geometric state $\mathcal{S}_{\text{geom}}$ determines the metric $g(t)$, nonlocal structure Ξ, and vacuum modulation $p(t)$:

$$\zeta_R(s, t) = \sum_k \lambda_k(t)^{-s} \tag{7}$$

The Rivero–Zeta function is a derived spectral invariant, not a fundamental field. Its role is diagnostic and constraining, not dynamical.

2. The Garkness Vacuum-Tension Geometry

2.1 The Garkness Field as a Vacuum-Tension Scalar

The field $G(x)$ is a vacuum-tension geometric scalar, not a particle field. The classical action for the Garkness field:

$$S_G = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu G \, \partial_\nu G - V(G) \right] \tag{8}$$

with potential:

$$V(G) = \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 \tag{9}$$

2.2 The Induced Dark-Matter Term

The Garkness field induces a dark-matter-like contribution:

$$\rho_{\rm DM} = \alpha_{\rm DM} G^2, \qquad p_{\rm DM} \approx 0 \tag{10}$$

This term is emergent from the vacuum-tension geometry, not postulated or added by hand.

2.3 Pause-Factor Vacuum Modulation

The vacuum energy is modulated by the pause-factor field $p(x)$:

$$\rho_{\Lambda,\rm eff} = p(x) \, \Lambda_{\rm eff} \tag{11}$$

with evolution:

$$\frac{dp}{dt} = -\gamma \big(p(t) - p_\infty \big), \qquad \gamma > 0 \tag{12}$$

2.4 Full Energy Density and Pressure

$$\rho_{\rm tot} = \rho_{\rm SM} + \frac{1}{2}\dot{G}^2 + \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 + \alpha_{\rm DM} G^2 + p(t) \Lambda_{\rm eff} \tag{13}$$

$$p_{\rm tot} = \frac{1}{2}\dot{G}^2 - \Lambda_G - \frac{1}{2} m_G^2 G^2 - \frac{\lambda_G}{4} G^4 - p(t) \Lambda_{\rm eff} \tag{14}$$

2.5 Field Equation for Garkness

Varying the action yields:

$$\nabla_\mu \nabla^\mu G + \big(m_G^2 + 2\alpha_{\rm DM} \big) G + \lambda_G G^3 = 0 \tag{15}$$

In FLRW background:

$$\ddot{G} + 3H\dot{G} + \big(m_G^2 + 2\alpha_{\rm DM} \big) G + \lambda_G G^3 = 0 \tag{16}$$

2.6 Friedmann System (Local Dynamics)

$$H^2 = \rho_{m0} \, a^{-3} + \frac{1}{2}\dot{G}^2 + \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 + \alpha_{\rm DM} G^2 + p(t) \Lambda_{\rm eff} \tag{17}$$

Acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{1}{2} \big(\rho_{\rm tot} + p_{\rm tot} \big) \tag{18}$$

3. The Ξ-Kernel and Nonlocal Correlation Geometry

3.1 Definition of the Ξ-Kernel

Definition 3.1 (Ξ-Kernel). $\Xi : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ *with properties:*

$$\textit{(i) Symmetry:} \quad \Xi(x,y) = \Xi(y,x) \tag{19}$$

$$\textit{(ii) Positivity:} \quad \int f(x)\,\Xi(x,y)\,f(y)\,d^4x\,d^4y \geq 0 \tag{20}$$

$$\textit{(iii) Integrability:} \quad \int_{\mathcal{M}} |\Xi(x,y)|\,d^4y < \infty \tag{21}$$

$$\textit{(iv) Covariance under diffeomorphisms} \tag{22}$$

3.2 Ξ as a Nonlocal Vacuum-Correlation Operator

$$(K_\Xi f)(x) = \int_{\mathcal{M}} \Xi(x,y)\,f(y)\,d^4y \tag{23}$$

$$G_\Xi(x) = (K_\Xi G)(x) = \int_{\mathcal{M}} \Xi(x,y)\,G(y)\,d^4y \tag{24}$$

3.3 Ξ-Modified Energy Functional

$$E_\Xi[G] = \frac{1}{2}\int d^4x \int d^4y\, G(x)\,\Xi(x,y)\,G(y) \tag{25}$$

$$E_{\text{Garkness}} = \int d^4x \left[\frac{1}{2}\dot{G}^2 + V(G)\right] + E_\Xi[G] \tag{26}$$

3.4 Ξ-Modified Field Equation

$$\nabla_\mu\nabla^\mu G + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 + \int_{\mathcal{M}} \Xi(x,y)\,G(y)\,d^4y = 0 \tag{27}$$

In FLRW:

$$\ddot{G} + 3H\dot{G} + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 + \int \Xi(t,t')\,G(t')\,dt' = 0 \tag{28}$$

3.5 Ξ and the Geometric Information Functional

$I_{\text{geom}} = I[G, \Xi, p; \sigma]$ with contributions:

$$I_1 = \int \Xi(x,y)\,G(x)\,G(y)\,d^4x\,d^4y \quad \text{(nonlocal correlation term)} \tag{29}$$

$$I_2 = \sum_{\sigma=1}^{33} w_\sigma \int_{U_\sigma} \Xi\,d^4x\,d^4y \quad \text{(sector weighting)} \tag{30}$$

$$I_3 = \int p(x)\left[\int \Xi(x,y)\,G(y)\,d^4y\right]d^4x \quad \text{(pause-factor coupling)} \tag{31}$$

3.6 Ξ and the 33-Web Sector Structure

$$\Xi_\sigma(x,y) = \Xi(x,y)\,\chi_\sigma(x)\,\chi_\sigma(y) \tag{32}$$

defines 33 geometric sectors, 33 correlation patterns, 33 asymptotic behaviors. The sector structure represents a block decomposition of correlation domains within a single universe.

3.7 Spectral Properties of Ξ

$$K_\Xi\phi_n = \lambda_n\phi_n, \qquad \lambda_n \geq 0 \tag{33}$$

4. The Geometric Information Functional I_{geom}

4.1 Definition of the Functional

$$I_{\text{geom}} : \mathcal{S}_{\text{geom}} \to \mathbb{R}, \qquad \mathcal{S}_{\text{geom}} = \big(G(x), \Xi(x,y), p(x); \sigma\big) \tag{34}$$

4.2 Structural Requirements

(i) Diffeomorphism covariance;

(ii) Positivity: $I_{\text{geom}} \geq 0$;

(iii) Sector consistency via sector-restricted integrals;

(iv) Finite variation: $\frac{\delta I_{\text{geom}}}{\delta G}$, $\frac{\delta I_{\text{geom}}}{\delta \Xi}$, $\frac{\delta I_{\text{geom}}}{\delta p}$ exist and are finite.

4.3 Decomposition of the Functional

$$I_{\text{geom}} = I_G + I_\Xi + I_p + I_\sigma \tag{35}$$

where:

$$I_G = \int_{\mathcal{M}} \left[\alpha_1 G^2 + \alpha_2 (\nabla G)^2 + \alpha_3 G^4\right] \sqrt{-g}\, d^4x \tag{36}$$

$$I_\Xi = \frac{1}{2} \int_{\mathcal{M}} \int_{\mathcal{M}} G(x)\, \Xi(x,y)\, G(y)\, d^4x\, d^4y \tag{37}$$

$$I_p = \beta_1 \int_{\mathcal{M}} p(x)\, G(x)^2 \sqrt{-g}\, d^4x + \beta_2 \int_{\mathcal{M}} p(x) \left[\int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y\right] d^4x \tag{38}$$

$$I_\sigma = \sum_{\sigma=1}^{33} w_\sigma \int_{U_\sigma} \left[G(x)^2 + \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y\right] d^4x \tag{39}$$

4.4 Variational Derivatives

$$\frac{\delta I_{\text{geom}}}{\delta G(x)} = 2\alpha_1 G(x) - 2\alpha_2 \nabla^2 G(x) + 4\alpha_3 G(x)^3 + \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y + \beta_1 p(x)\, G(x) + \beta_2 \int_{\mathcal{M}} p(y)\, \Xi(x,y)\, d^4y + \text{sector terms} \tag{40}$$

$$\frac{\delta I_{\text{geom}}}{\delta \Xi(x,y)} = \frac{1}{2}\, G(x)\, G(y) + \beta_2\, p(x)\, G(y) + \text{sector terms} \tag{41}$$

$$\frac{\delta I_{\text{geom}}}{\delta p(x)} = \beta_1\, G(x)^2 + \beta_2 \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y \tag{42}$$

4.5 Information-Modulated Cosmological Dynamics

Local FLRW + Scalar + Pause subject to:

$$\frac{\delta I_{\text{geom}}}{\delta G} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta \Xi} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta p} = 0 \tag{43}$$

5. Information-Modulated FLRW Cosmology

5.1 The Full Dynamical System

(i) Friedmann equation:

$$H^2 = \rho_{m0}\, a^{-3} + \frac{1}{2}\dot{G}^2 + \Lambda_G + \frac{1}{2}m_G^2 G^2 + \frac{\lambda_G}{4}G^4 + \alpha_{\rm DM} G^2 + p(t)\Lambda_{\rm eff} + \rho_\Xi[G] \tag{44}$$

where:

$$\rho_\Xi[G] = \frac{1}{2a^3}\int G(t)\,\Xi(t,t')\,G(t')\,dt' \tag{45}$$

(ii) Garkness field equation (Ξ-extended):

$$\ddot{G} + 3H\dot{G} + \big(m_G^2 + 2\alpha_{\rm DM}\big)G + \lambda_G G^3 + \int \Xi(t,t')\,G(t')\,dt' = -\frac{\delta I_{\rm geom}}{\delta G(t)} \tag{46}$$

(iii) Pause-factor evolution:

$$\frac{dp}{dt} = -\gamma\big(p(t) - p_\infty\big) - \frac{\delta I_{\rm geom}}{\delta p(t)} \tag{47}$$

(iv) Ξ-kernel consistency:

$$\frac{\delta I_{\rm geom}}{\delta \Xi(t,t')} = 0 \tag{48}$$

(v) Scale factor evolution:

$$\frac{da}{dt} = H\,a \tag{49}$$

5.2 Three-Layer Cosmology

Layer 1 – Local Dynamics: $H(t)$, $a(t)$, matter dilution a^{-3}.

Layer 2 – Vacuum-Tension: $G(t)$, $\alpha_{\rm DM}G^2$, $V(G)$, $p(t)\Lambda_{\rm eff}$.

Layer 3 – Information Geometry: nonlocal correlations, geometric constraints, sector structure.

5.3 Effective Equation of State

$$w_{\rm eff}(t) = \frac{p_{\rm tot}(t)}{\rho_{\rm tot}(t)} \tag{50}$$

$$\rho_{\rm tot} = \rho_{m0}\, a^{-3} + \rho_G + \rho_{\rm DM} + \rho_{\Lambda,\rm eff} + \rho_\Xi \tag{51}$$

5.4 Deceleration Parameter

$$q(t) = \frac{1}{2}\big(1 + 3w_{\rm eff}(t)\big) \tag{52}$$

Acceleration occurs when $w_{\rm eff} < -\frac{1}{3}$. Late-time attractor: $w_{\rm eff} \to -1$, $q \to -1$.

5.5 The Information-Modulated Attractor

Fixed point:

$$\big(G, \dot{G}, p\big) \to \big(G^*, 0, p_\infty\big), \qquad H \to H_\infty > 0 \tag{53}$$

selected by $\delta I_{\rm geom} = 0$.

6. Spectral Geometry and the Rivero–Zeta in the Garkness Background

6.1 The Spectral Operator in a Garkness Background

$$\mathcal{D}(t) = -\Delta_{g(t)} + M^2 + U_G(t) + U_\Xi(t) \tag{54}$$

where:

$$U_G(t) = m_G^2\, G(t)^2 + \lambda_G\, G(t)^4 \tag{55}$$

$$U_\Xi(t) = \int \Xi(t,t')\, G(t')\, dt' \tag{56}$$

The spectral operator $\mathcal{D}(t)$ depends smoothly on the background geometry.

6.2 Eigenvalue Problem

$$\mathcal{D}(t)\, \phi_k(t,x) = \lambda_k(t)\, \phi_k(t,x) \tag{57}$$

For flat FLRW: $\phi_k = e^{i\mathbf{k}\cdot\mathbf{x}}$, $\mathbf{k} \in \mathbb{R}^3$:

$$\lambda_k(t) = \frac{k^2}{a(t)^2} + M^2 + U_G(t) + U_\Xi(t) \tag{58}$$

6.3 The Rivero–Zeta Function

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s} \tag{59}$$

In continuum:

$$\zeta_R(s,t) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \left[\frac{k^2}{a(t)^2} + M^2 + U_G(t) + U_\Xi(t) \right]^{-s} d^3k \tag{60}$$

Converges for $\mathrm{Re}(s) > \frac{3}{2}$.

The Rivero–Zeta function $\zeta_R(s,t)$ is a derived spectral invariant, not a fundamental field. Its role is diagnostic and constraining, not dynamical.

6.4 Analytic Continuation

Admits analytic continuation to the entire complex plane except simple poles at $s = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, \ldots$

6.5 Spectral Determinant

$$\det \mathcal{D}(t) = \exp\left[-\zeta_R'(0,t)\right] \tag{61}$$

6.6 Spectral Flow Equation

$$\frac{d}{dt}\zeta_R(s,t) = -s \sum_k \lambda_k(t)^{-s-1}\, \frac{d\lambda_k}{dt} \tag{62}$$

where:

$$\frac{d\lambda_k}{dt} = -2H(t)\, \frac{k^2}{a(t)^2} + \frac{dU_G}{dt} + \frac{dU_\Xi}{dt} \tag{63}$$

6.7 Rivero–Zeta as a Constraint on Cosmology

$$\frac{d}{dt}\zeta_R(0,t) = 0 \tag{64}$$

yields the spectral consistency condition coupling expansion, vacuum tension, nonlocal correlations, and information geometry.

6.8 Integration with Information Functional

I_{geom} contains $\gamma_R\,\zeta_R(s_0,t)$. All variational derivatives acquire spectral contributions.

7. Sector Structure and the 33-Web Geometry

7.1 Sector Decomposition of the Manifold

$$\mathcal{M} = \bigcup_{\sigma=1}^{33} U_\sigma, \qquad U_\sigma \cap U_{\sigma'} = \varnothing \quad \text{for } \sigma \neq \sigma' \tag{65}$$

Indicator functions:

$$\chi_\sigma(x) = \begin{cases} 1 & \text{if } x \in U_\sigma \\ 0 & \text{otherwise} \end{cases} \tag{66}$$

The sector labels σ are solution-space decomposition indices, not physical universes. The 33-web is a block decomposition of the Ξ-kernel and a partition of correlation structure inside a single universe.

7.2 Sector-Restricted Fields

$$G_\sigma(x) = \chi_\sigma(x)\, G(x), \qquad p_\sigma(x) = \chi_\sigma(x)\, p(x) \tag{67}$$

7.3 Sector-Restricted Ξ-Kernel

$$\Xi(x,y) = \sum_{\sigma,\sigma'=1}^{33} \Xi_{\sigma\sigma'}(x,y) \tag{68}$$

where:

$$\Xi_{\sigma\sigma'}(x,y) = \chi_\sigma(x)\, \Xi(x,y)\, \chi_{\sigma'}(y) \tag{69}$$

Diagonal blocks $\Xi_{\sigma\sigma}$ encode intra-sector correlations; off-diagonal $\Xi_{\sigma\sigma'}$ encode inter-sector correlations.

7.4 Sector-Dependent Information Content

$$I_\sigma = w_\sigma \int_{U_\sigma} \left[G(x)^2 + \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y \right] d^4x \tag{70}$$

$$I_{\text{geom}} = \sum_{\sigma=1}^{33} I_\sigma + (\text{local} + \text{nonlocal} + \text{modulation terms}) \tag{71}$$

7.5 Sector-Dependent Spectral Structure

$$\mathcal{D}(t) = \bigoplus_{\sigma=1}^{33} \mathcal{D}_\sigma(t) + \text{inter-sector couplings} \tag{72}$$

$$\zeta_R(s,t) = \sum_{\sigma=1}^{33} \zeta_R^{(\sigma)}(s,t) \tag{73}$$

7.6 Sector-Dependent Dynamics

$$\ddot{G}_\sigma + 3H\dot{G}_\sigma + \left(m_G^2 + 2\alpha_{\text{DM}}\right) G_\sigma + \lambda_G G_\sigma^3 + \int \Xi_{\sigma\sigma'}(t,t')\, G_{\sigma'}(t')\, dt' = -\frac{\delta I_{\text{geom}}}{\delta G_\sigma} \tag{74}$$

7.7 Sector-Dependent Attractors

Each sector:

$$(G_\sigma, \dot{G}_\sigma, p_\sigma) \to (G_{\sigma,*}, 0, p_{\infty,\sigma}) \tag{75}$$

Global:

$$(G, \dot{G}, p) \to \left(\sum_{\sigma=1}^{33} G_{\sigma,*},\ 0,\ \sum_{\sigma=1}^{33} p_{\infty,\sigma} \right) \tag{76}$$

8. The Full AI33-MPOPT Master Equation

8.1 The Unified Variational Principle

$$S_{\text{AI33}} = S_{\text{local}} + S_{\Xi} + S_{\text{spec}} + I_{\text{geom}} \tag{77}$$

Master equation obtained by:

$$\delta S_{\text{AI33}} = 0 \tag{78}$$

8.2 The Master Equation (Unified Form)

$$\mathcal{D}_{\text{local}}[G,p,a] + \mathcal{D}_{\Xi}[G] + \mathcal{D}_{\text{spec}}[G,\Xi,p] + \frac{\delta I_{\text{geom}}}{\delta G} = 0 \tag{79}$$

Companion equations:

$$\frac{\delta I_{\text{geom}}}{\delta p} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta \Xi} = 0, \qquad \frac{d}{dt}\zeta_R(0,t) = 0 \tag{80}$$

8.3 Local Dynamics Contribution

$$\mathcal{D}_{\text{local}}[G,p,a] = \ddot{G} + 3H\dot{G} + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 \tag{81}$$

8.4 Nonlocal Ξ-Kernel Contribution

$$\mathcal{D}_{\Xi}[G] = \int \Xi(t,t')\, G(t')\, dt' \tag{82}$$

8.5 Spectral Contribution (Rivero–Zeta)

$$\mathcal{D}_{\text{spec}} = -\gamma_R\, s_0 \sum_k \lambda_k(t)^{-s_0-1}\, \frac{\partial \lambda_k(t)}{\partial G} \tag{83}$$

8.6 Information-Functional Contribution

$$\frac{\delta I_{\text{geom}}}{\delta G} = 2\alpha_1 G - 2\alpha_2 \nabla^2 G + 4\alpha_3 G^3 + \int \Xi\, G + \beta_1 p\, G + \beta_2 \int p\, \Xi + \text{sector terms} \tag{84}$$

8.7 Sectoral Contribution (33-Web)

$$\frac{\delta I_{\text{geom}}}{\delta G_\sigma} = w_\sigma \left[2G_\sigma + \int \Xi_{\sigma\sigma'}\, G_{\sigma'}\, dt'\right] \tag{85}$$

8.8 The Fully Expanded Master Equation

$$\begin{aligned} &\ddot{G} + 3H\dot{G} + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 + \int \Xi(t,t')\, G(t')\, dt' \\ &- \gamma_R\, s_0 \sum_k \lambda_k(t)^{-s_0-1}\, \frac{\partial \lambda_k}{\partial G} + 2\alpha_1 G - 2\alpha_2 \nabla^2 G + 4\alpha_3 G^3 \\ &+ \beta_1 p\, G + \beta_2 \int p(t')\, \Xi(t,t')\, dt' + \sum_{\sigma=1}^{33} w_\sigma \left[2G_\sigma + \int \Xi_{\sigma\sigma'}\, G_{\sigma'}\, dt'\right] = 0 \end{aligned} \tag{86}$$

This is the **AI33-MPOPT Master Equation** encoding local, nonlocal, spectral, informational, and sectoral physics.

9. Cosmological Solutions and Attractor Structure

9.1 Dynamical Variables

$a(t)$, $H(t)$, $G(t)$, $p(t)$, $\Xi(t,t')$, σ, $\lambda_k(t)$.

Evolution equations: local FLRW, Ξ-extended Garkness, pause-factor modulation, spectral flow, sector constraints, information constraints.

9.2 Fixed Points of the System

Fixed point conditions: $\dot{a}=0$, $\dot{H}=0$, $\dot{G}=0$, $\dot{p}=0$, $\partial_t\Xi=0$, $\partial_t\lambda_k=0$.

$$\text{(i) } G(t)\to G^* \tag{87}$$
$$\text{(ii) } p(t)\to p_\infty \tag{88}$$
$$\text{(iii) } U_\Xi(t)\to U_{\Xi,*} \tag{89}$$
$$\text{(iv) } \lambda_k(t)\to\lambda_{k,*} \tag{90}$$
$$\text{(v) } H(t)\to H_\infty>0 \tag{91}$$

This is a de Sitter-like attractor with information-modulated structure.

9.3 Fixed-Point Equation for G^*

$$\big(m_G^2+2\alpha_{\rm DM}\big)G^*+\lambda_G(G^*)^3+\int\Xi^*(t,t')\,G^*\,dt'+\frac{\partial}{\partial G}I_{\rm geom}(G^*)=0 \tag{92}$$

9.4 Fixed-Point Hubble Parameter

$$H_\infty^2=\Lambda_G+\frac{1}{2}m_G^2(G^*)^2+\frac{\lambda_G}{4}(G^*)^4+\alpha_{\rm DM}(G^*)^2+p_\infty\Lambda_{\rm eff}+\rho_{\Xi,*} \tag{93}$$

9.5 Stability Analysis

Linearize: $G(t)=G^*+\delta G(t)$, $p(t)=p_\infty+\delta p(t)$, $H(t)=H_\infty+\delta H(t)$.

Stability criterion:

$$m_{\rm eff}^2+\frac{\partial^2 I_{\rm geom}}{\partial G^2}+\lambda_{\Xi,\min}>0 \tag{94}$$

where:

$$m_{\rm eff}^2=m_G^2+2\alpha_{\rm DM}+3\lambda_G(G^*)^2 \tag{95}$$

9.6 Sector-Dependent Stability

Each sector σ:

$$m_{{\rm eff},\sigma}^2+\frac{\partial^2 I_{\rm geom}}{\partial G_\sigma^2}+\lambda_{\Xi,\sigma,\min}>0 \tag{96}$$

(Assumption; proof deferred to future work.)

9.7 Cosmological Phases

Phase I – Matter-Dominated: $\rho_{m0}\,a^{-3}\gg\rho_{\rm dark}$, $a(t)\sim t^{2/3}$.

Phase II – Information-Modulated Transition: $\rho_{\rm dark}\sim\rho_{m0}\,a^{-3}$, nonlocal correlations grow, spectral flow accelerates.

Phase III – Information-Selected Attractor: $\delta I_{\rm geom}=0$, $H\to H_\infty$, $w_{\rm eff}\to-1$, $q\to-1$.

9.8 Observational Predictions

(i) Non-monotonic $w_{\text{eff}}(z)$ with structure from $G(z)$, $p(z)$, $\Xi(z)$, $\zeta_R(z)$.

(ii) Modified transition redshift $z_{\text{trans}} \neq z_{\Lambda\text{CDM}}$.

(iii) Sector-dependent signatures: anisotropic imprints, spectral distortions, correlation anomalies.

(iv) Spectral-geometric signature: Rivero–Zeta evolves differently from ΛCDM.

10. Physical Interpretation and Conceptual Architecture

10.1 Local Dynamics: The Observable Universe

FLRW metric, $a(t)$, $H(t)$, $\rho_{m0}\, a^{-3}$. Constrained by supernova data, BAO, CMB, structure formation.

10.2 Vacuum-Tension Geometry: The Garkness Field

$G(x)$ is a vacuum-tension geometric scalar, not a particle field. It contributes: scalar potential $V(G)$, induced dark-matter $\alpha_{\mathrm{DM}} G^2$ (emergent, not postulated), vacuum-modulated term $p(t)\Lambda_{\mathrm{eff}}$.

10.3 Nonlocal Correlation Geometry: The Ξ-Kernel

Ξ is a symmetric, positive, integrable nonlocal operator encoding long-range vacuum structure and geometric correlations. It is non-signaling and causally safe. It acts as a geometric correlation constraint, not a propagating field.

10.4 Geometric Information: The Functional I_{geom}

The functional I_{geom} is a constraint selector, not a dynamical action. It encodes: local vacuum structure, nonlocal correlations, pause-factor modulation, sector structure, spectral contributions. It represents global consistency conditions and the late-time attractor mechanism.

10.5 Spectral Geometry: The Rivero–Zeta Layer

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s} \tag{97}$$

The Rivero–Zeta function is a derived spectral invariant, not a fundamental field. Its role is diagnostic and constraining, not dynamical. The spectral operator depends smoothly on the background geometry.

10.6 Sector Structure: The 33-Web

Sector labels σ are solution-space decomposition indices, not physical universes. The 33-web is a block decomposition of the Ξ-kernel and a partition of correlation structure inside a single universe.

10.7 How the Layers Interact

(i) Local $\to$ Vacuum-Tension: $H(t)$ damps $G(t)$
(ii) Vacuum-Tension $\to$ Local: G contributes to energy density
(iii) Nonlocal $\to$ Vacuum-Tension: Ξ modifies effective potential
(iv) Information $\to$ All: I_{geom} imposes global constraints
(v) Spectral $\to$ Information: ζ_R contributes to I_{geom}
(vi) Sectoral $\to$ Nonlocal: 33-web determines Ξ block structure (98)

10.8 Physical Picture of AI33-MPOPT Universe

Locally FLRW, globally constrained by geometric information, modulated by vacuum-tension, structured by nonlocal correlations, characterized by spectral invariants, organized into 33 geometric sectors.

11. Observational Signatures and Testable Predictions

11.1 Effective Equation of State $w_{\text{eff}}(z)$

$$w_{\text{eff}}(z) = \frac{p_{\text{tot}}(z)}{\rho_{\text{tot}}(z)} \tag{99}$$

Predictions: (i) Non-monotonic behavior unlike ΛCDM ($w = -1$), (ii) Smoother matter→dark transition, (iii) Shifted transition redshift $z_{\text{trans}}^{\text{AI33}} \neq z_{\text{trans}}^{\Lambda\text{CDM}}$. Testable with supernova and BAO data.

11.2 Modified Hubble Expansion $H(z)$

$$H^2(z) = \rho_{m0}(1+z)^3 + \rho_G(z) + \rho_{\text{DM}}(z) + \rho_{\Lambda,\text{eff}}(z) + \rho_\Xi(z) \tag{100}$$

Predictions: (i) Mild early-dark-energy-like effect, (ii) Late-time spectral correction, (iii) Smoother approach to de Sitter.

11.3 Growth of Structure $f\sigma_8(z)$

Predictions: (i) Scale-dependent growth suppression from Ξ-kernel nonlocal damping, (ii) Mild deviation from ΛCDM at $z < 1$. Testable with RSD, weak lensing, galaxy clustering.

11.4 CMB Signatures

Predictions: (i) Modified ISW signal from smoother $w_{\text{eff}}(z)$ transition, (ii) Sector-dependent correlation anomalies.

11.5 Spectral Signatures (Rivero–Zeta)

Predictions: (i) Time-dependent spectral flow $\frac{d}{dt}\zeta_R(s,t) \neq 0$ during transition, (ii) Late-time spectral freeze as attractor reached, (iii) Sector-resolved spectral fingerprint. (Assumption; quantitative predictions deferred to future work.)

11.6 Nonlocal Correlation Signatures

Ξ-kernel produces: (i) Long-range correlation distortions in lensing/galaxy correlations/CMB, (ii) Scale-dependent deviations in $P(k)$, (iii) Sector-dependent anisotropies.

11.7 Summary of Testable Predictions

Non-monotonic $w_{\text{eff}}(z)$, smoother transition, modified Hubble, scale-dependent growth suppression, ISW anomalies, spectral flow signatures, sector correlation patterns. Quantitative, falsifiable, distinct from ΛCDM.

12. Mathematical Consistency and Theorem Structure

12.1 Preliminaries and Functional Spaces

$(\mathcal{M}, g) = (\mathbb{R}^3 \times \mathbb{R}, -dt^2 + a(t)^2 d\mathbf{x}^2)$.

Variables: $G(t) \in C^2(\mathbb{R})$, $p(t) \in C^1(\mathbb{R})$, $\Xi(t,t') \in L^2(\mathbb{R}^2)$, $a(t) \in C^2(\mathbb{R})$, $\lambda_k(t) \in C^1(\mathbb{R})$.

I_{geom}: Fréchet differentiable, bounded below, coercive.

Theorem 12.1 (Well-posedness). *System*

$$\begin{aligned} &\ddot{G} + 3H\dot{G} + V'(G) = F_\Xi[G] + F_I[G,p] \\ &\dot{p} = -\gamma(p - p_\infty) - \frac{\delta I_{\text{geom}}}{\delta p} \\ &H^2 = \rho_{\text{tot}} \end{aligned} \tag{101}$$

admits unique local solution for admissible initial data. (Picard–Lindelöf)

Theorem 12.2 (Energy Boundedness). *If $\lambda_G > 0$ and Ξ positive semidefinite, then:*

$$E_{\text{tot}} = E_G + E_\Xi + E_p + E_{\text{spec}} \quad \textit{is bounded below} \tag{102}$$

Theorem 12.3 (Existence of Fixed Points). *There exists at least one fixed point $(G^*, p_\infty, H_\infty, \Xi^*, \lambda_{k,*})$ satisfying the master equation.*

Theorem 12.4 (Stability of Attractor). *If:*

$$M_{\text{eff}} = m_G^2 + 2\alpha_{\text{DM}} + 3\lambda_G (G^*)^2 + \left.\frac{\partial^2 I_{\text{geom}}}{\partial G^2}\right|_{G^*} + \lambda_{\Xi,\min} > 0 \tag{103}$$

then the fixed point is asymptotically stable.

Theorem 12.5 (Spectral Continuity). *Eigenvalues $\lambda_k(t)$ of $\mathcal{D}(t)$ depend continuously on t. (Kato perturbation theory)*

Theorem 12.6 (Sector Consistency). *The 33-sector decomposition is preserved: $\Xi_{\sigma\sigma'}(t_0, t_0') = 0$ for $\sigma \neq \sigma'$ implies $\Xi_{\sigma\sigma'}(t,t') = 0$ for all t. The 33-web is dynamically stable.*

13. Synthesis: The AI33-MPOPT Unified Architecture

13.1 The Six-Layer Structure

1. **Local Cosmological Dynamics:** FLRW, Hubble, matter dilution.
2. **Vacuum-Tension Geometry (Garkness):** scalar vacuum, dark-matter, pause-factor.
3. **Nonlocal Correlation Geometry (Ξ):** long-range correlations, geometric constraints.
4. **Geometric Information Functional:** global constraints, constraint selection.
5. **Spectral Geometry (Rivero–Zeta):** spectral operator, eigenvalues, determinant.
6. **Sector Structure (33-Web):** decomposition, sector dynamics, sector spectra.

13.2 Architecture as Directed Graph

Local $\to$ Vacuum-Tension $\to$ Nonlocal $\to$ Information $\leftarrow$ Spectral, all organized by 33-Sector Structure.

13.3 Master Equation as Central Node

$$\mathcal{D}_{\text{local}} + \mathcal{D}_{\Xi} + \mathcal{D}_{\text{spec}} + \frac{\delta I_{\text{geom}}}{\delta G} = 0 \tag{104}$$

receives input from all layers, outputs universe evolution, determines attractor, enforces consistency.

13.4 Attractor as Architectural Endpoint

$$(G, \dot{G}, p) \to (G^*, 0, p_\infty), \quad H \to H_\infty, \quad \lambda_k \to \lambda_{k,*}, \quad \Xi \to \Xi^* \tag{105}$$

Information-selected, spectrally consistent, sector-resolved, stable, globally constrained vacuum state.

13.5 Architecture as Hierarchical Stack

Layer 1: Metric. Layer 2: Vacuum-Tension. Layer 3: Nonlocal. Layer 4: Information. Layer 5: Spectral. Layer 6: Sector.

13.6 Architecture as Research Program

Paper I: Reduced FLRW. Paper II: Rivero–Zeta. Paper III: Information-Modulated Cosmology. Paper IV: Nonlocal Geometry. Paper V: Sector Structure. Paper VI: Master Equation. Paper VII: Observational Signatures.

14. Future Extensions and Open Directions

14.1 Extension A – Full Nonlocal Dynamics of Ξ-Kernel

Dynamical theory:

$$\partial_t \Xi(t, t') = F[G, \Xi, p; \sigma] \tag{106}$$

with symmetry, positivity, sector-block, spectral preservation. (Assumption; proof deferred to future work.)

14.2 Extension B – Covariant Generalization Beyond FLRW

Generalize to: Bianchi I/IX, perturbed FLRW, full 3+1 ADM.

14.3 Extension C – Perturbation Theory and Structure Formation

Full perturbation theory: scalar, vector, tensor perturbations with nonlocal Ξ-correlations, information-modulated vacuum tension, sector-dependent dynamics.

14.4 Extension D – Spectral Flow and Renormalization

Spectral renormalization group:

$$\frac{d}{d \ln k} \lambda_k(t) = R[G, \Xi, p; \sigma] \tag{107}$$

14.5 Extension E – Sector Dynamics and Phase Transitions

Sector transitions, bifurcations, mergers governed by $\Delta\sigma = \mathcal{G}[G, \Xi, p; \sigma]$.

14.6 Extension F – Information-Theoretic Interpretation

Extend I_{geom} to include: entropy-like terms, Fisher information, KL divergences.

14.7 Extension G – Numerical Simulation Program

Simulate: master equation, spectral flow, Ξ-kernel evolution, sector dynamics.

14.8 Extension H – Observational Confrontation

Fit $w_{\text{eff}}(z)$ to supernovae, compute BAO distances, predict ISW, compute $f\sigma_8(z)$.

15. Conclusion of the Master File

15.1 Summary of Architectural Layers

Six interacting layers: Local Dynamics (FLRW backbone), Vacuum-Tension Geometry (Garkness field), Nonlocal Correlation Geometry (Ξ-kernel), Geometric Information Functional (global constraints), Spectral Geometry (Rivero–Zeta), Sector Structure (33-web).

15.2 The Unified Master Equation

$$\mathcal{D}_{\text{local}} + \mathcal{D}_{\Xi} + \mathcal{D}_{\text{spec}} + \frac{\delta I_{\text{geom}}}{\delta G} = 0 \tag{108}$$

with:

$$\frac{\delta I_{\text{geom}}}{\delta p} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta \Xi} = 0, \qquad \frac{d}{dt}\zeta_R(0,t) = 0 \tag{109}$$

15.3 The Information-Selected Attractor

$$(G, \dot{G}, p) \to (G^*, 0, p_\infty), \quad H \to H_\infty, \quad \lambda_k \to \lambda_{k,*}, \quad \Xi \to \Xi^* \tag{110}$$

Dynamically stable, spectrally consistent, sector-resolved, globally constrained.

15.4 Observational Consequences

Non-monotonic $w_{\text{eff}}(z)$, smoother transition, modified Hubble, scale-dependent growth suppression, ISW anomalies, sector correlation patterns, spectral flow signatures.

15.5 Mathematical Consistency

Well-posedness, energy boundedness, fixed point existence, attractor stability, spectral continuity, sector preservation.

15.6 Conceptual Unity

Synthesis of: geometry (metric, vacuum tension, nonlocal structure), information (global constraints, modulation), spectral theory (Rivero–Zeta), dynamics (master equation, attractors), structure (33-web).

15.7 Closing Statement

AI33-MPOPT provides: new geometric foundation for cosmology, mathematically controlled ΛCDM extension, unified local/nonlocal/spectral/informational treatment, structured extensible architecture. This master file completes the foundational construction of the theory.

A. Notation and Definitions

A.1 Spacetime and Geometry

$(\mathcal{M}, g)$: 4D Lorentzian manifold, signature $(-,+,+,+)$.
FLRW: $ds^2 = -dt^2 + a(t)^2\, d\mathbf{x}^2$.
Hubble: $H(t) = \dot{a}(t)/a(t)$.
Laplace–Beltrami: $\Delta_{g(t)}$.

A.2 Fields and Variables

$G : \mathcal{M} \to \mathbb{R}$ (Garkness vacuum-tension scalar), $p : \mathcal{M} \to \mathbb{R}$ (pause-factor), $\Xi : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ (nonlocal kernel), $\sigma \in \{1, \ldots, 33\}$ (sector index), $\lambda_k(t) \in \mathbb{R}_+$ (eigenvalues).

A.3 Potentials and Energies

$$V(G) = \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 \tag{111}$$

$$\rho_{\mathrm{DM}} = \alpha_{\mathrm{DM}} G^2 \quad \text{(emergent)} \tag{112}$$

$$\rho_{\Lambda,\mathrm{eff}} = p(t)\Lambda_{\mathrm{eff}} \tag{113}$$

$$\rho_\Xi = \frac{1}{2a^3} \int G\, \Xi\, G\, dt' \tag{114}$$

A.4 Operators and Functionals

$$(K_\Xi f)(x) = \int_{\mathcal{M}} \Xi(x,y)\, f(y)\, d^4y \tag{115}$$

$$\mathcal{D}(t) = -\Delta_g + M^2 + U_G + U_\Xi \tag{116}$$

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s} \tag{117}$$

$$I_{\mathrm{geom}} = I_G + I_\Xi + I_p + I_\sigma \tag{118}$$

A.5 Sector Decomposition

$$\mathcal{M} = \bigcup_{\sigma=1}^{33} U_\sigma \tag{119}$$

$$\chi_\sigma(x) = \begin{cases} 1 & x \in U_\sigma \\ 0 & \text{otherwise} \end{cases} \tag{120}$$

$$G_\sigma = \chi_\sigma G \tag{121}$$

$$\Xi_{\sigma\sigma'} = \chi_\sigma\, \Xi\, \chi_{\sigma'} \tag{122}$$

A.6 Dynamical Equations

Garkness:

$$\ddot{G} + 3H\dot{G} + (m_G^2 + 2\alpha_{\mathrm{DM}})G + \lambda_G G^3 + \int \Xi\, G\, dt' = -\frac{\delta I_{\mathrm{geom}}}{\delta G} \tag{123}$$

Pause:

$$\dot{p} = -\gamma(p - p_\infty) - \frac{\delta I_{\mathrm{geom}}}{\delta p} \tag{124}$$

Spectral:

$$\frac{d}{dt}\zeta_R(0,t) = 0 \tag{125}$$

A.7 Cosmological Quantities

$$\rho_{\rm tot} = \rho_{m0}\, a^{-3} + \rho_G + \rho_{\rm DM} + \rho_{\Lambda,\rm eff} + \rho_\Xi \tag{126}$$

$$p_{\rm tot} = p_G - p(t)\Lambda_{\rm eff} \tag{127}$$

$$w_{\rm eff} = p_{\rm tot}/\rho_{\rm tot} \tag{128}$$

$$q = \frac{1}{2}(1 + 3w_{\rm eff}) \tag{129}$$

B. Derivations of Key Equations

B.1 Garkness Field Equation

From $S_{\text{local}} = \int d^4x\,\sqrt{-g}\,\left[\frac{1}{2}\dot{G}^2 - V(G)\right]$.

Variation of kinetic term:

$$\delta S_{\text{kin}} = -\int a^3\left[\ddot{G} + 3H\dot{G}\right]\delta G\,dt \tag{130}$$

Variation of potential:

$$\delta S_{\text{pot}} = -\int a^3\,V'(G)\,\delta G\,dt \tag{131}$$

Setting $\delta S = 0$:

$$\ddot{G} + 3H\dot{G} + m_G^2 G + \lambda_G G^3 = 0 \tag{132}$$

B.2 Nonlocal Ξ Contribution

From $S_\Xi = \frac{1}{2}\iint G\,\Xi\,G\,d^4x\,d^4y$.

Variation:

$$\frac{\delta S_\Xi}{\delta G(x)} = \int_{\mathcal{M}} \Xi(x,y)\,G(y)\,d^4y = \mathcal{D}_\Xi[G] \tag{133}$$

B.3 Pause-Factor Equation

Relaxation term:

$$\delta S_{\text{relax}} = -\int a^3\,\gamma(p - p_\infty)\,\delta p\,dt \tag{134}$$

Information contribution:

$$\frac{\delta I_p}{\delta p} = \beta_1 G^2 + \beta_2 \int \Xi\,G \tag{135}$$

Result:

$$\dot{p} = -\gamma(p - p_\infty) - \frac{\delta I_{\text{geom}}}{\delta p} \tag{136}$$

B.4 Friedmann Equation

$$\rho_{\text{tot}} = \rho_{m0}\,a^{-3} + \rho_G + \rho_{\text{DM}} + \rho_{\Lambda,\text{eff}} + \rho_\Xi \tag{137}$$

where $\rho_G = \frac{1}{2}\dot{G}^2 + V(G)$.

Friedmann:

$$H^2 = \rho_{\text{tot}} \tag{138}$$

B.5 Spectral Flow

$$\frac{d}{dt}\zeta_R(s,t) = -s\sum_k \lambda_k^{-s-1}\,\dot{\lambda}_k \tag{139}$$

where:

$$\dot{\lambda}_k = \langle \phi_k, \dot{\mathcal{D}}\,\phi_k\rangle = -2H\,\frac{k^2}{a^2} + \dot{U}_G + \dot{U}_\Xi \tag{140}$$

C. Spectral Operator Details

C.1 Definition

$$\mathcal{D}(t) = -\Delta_{g(t)} + M^2 + U_G(t) + U_\Xi(t) \tag{141}$$

where:

$$U_G = m_G^2 G^2 + \lambda_G G^4 \tag{142}$$

$$U_\Xi = \int \Xi(t,t')\, G(t')\, dt' \tag{143}$$

C.2 Self-Adjointness

Domain: $\mathcal{D}(\mathcal{D}(t)) = H^2(\mathbb{R}^3)$.
Positivity: $\langle f, \mathcal{D}(t) f \rangle \geq M^2 \|f\|^2$.

C.3 Eigenvalues

$\mathcal{D}(t)\,\phi_k = \lambda_k(t)\,\phi_k$.
For flat FLRW: $\phi_k = e^{i\mathbf{k}\cdot\mathbf{x}}$:

$$\lambda_k(t) = \frac{k^2}{a(t)^2} + M^2 + U_G(t) + U_\Xi(t) \tag{144}$$

C.4 Rivero–Zeta

$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s}$ converges for $\mathrm{Re}(s) > \frac{3}{2}$.
Analytic continuation with poles at $s = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, \ldots$
Determinant:

$$\det \mathcal{D}(t) = \exp\left[-\zeta_R'(0,t)\right] \tag{145}$$

C.5 Spectral Flow and Heat Kernel

Flow:

$$\frac{d}{dt}\zeta_R = -s \sum_k \lambda_k^{-s-1} \left[-2H\,\frac{k^2}{a^2} + \dot{U}_G + \dot{U}_\Xi\right] \tag{146}$$

Heat kernel:

$$\zeta_R(s,t) = \frac{1}{\Gamma(s)} \int_0^\infty \tau^{s-1}\, \mathrm{Tr}\left(e^{-\tau \mathcal{D}}\right) d\tau \tag{147}$$

D. Sector Decomposition Proofs

Theorem D.1 (Orthogonality). *For $\sigma \neq \sigma'$:* $\mathrm{supp}(G_\sigma) \cap \mathrm{supp}(G_{\sigma'}) = \varnothing$.

Proof. $\chi_\sigma \chi_{\sigma'} = 0 \Rightarrow G_\sigma G_{\sigma'} = 0$. □

Theorem D.2 (Block Structure). $\Xi = \sum_{\sigma,\sigma'} \Xi_{\sigma\sigma'}$ *with* $\Xi_{\sigma\sigma'} = 0$ *unless* $x \in U_\sigma$, $y \in U_{\sigma'}$.

Theorem D.3 (Preservation). $\Xi_{\sigma\sigma'}(t_0, t_0') = 0$ *for* $\sigma \neq \sigma'$ *implies* $\Xi_{\sigma\sigma'}(t, t') = 0$ *for all* t.

Proof. $\frac{\delta I_{\mathrm{geom}}}{\delta \Xi_{\sigma\sigma'}} = 0$ identically, so $\partial_t \Xi_{\sigma\sigma'} = 0$. □

Theorem D.4 (Sector Decomposition of Dynamics). *Sector dynamics decomposes into 33 independent equations.*

Theorem D.5 (Spectral Decomposition).

$$\mathcal{D}(t) = \bigoplus_{\sigma=1}^{33} \mathcal{D}_\sigma(t) \tag{148}$$

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